

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/ CE/EE/IT)

SEM - II University Examination, April- 2013

MA - 102 Engineering Mathematics-II

Date: 29.04.13, Monday

Time: 1.30 p. m to 4.30 p. m.

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks.

SECTION-I

Q-1 (a) Solve by the method of variation of parameters : $\frac{d^2y}{dx^2} + 16y = \tan 4x$. [05]

(b) Solve $\frac{dy}{dx} + 2y \tan x = \sin x$ [02]

Q-2 Solve the following differential equations

(a) $\frac{dy}{dx} = 2y \tan x + y^2 \tan^2 x$, if $y(0) = 1$ [04]

(b) $y''' + 3y'' + 3y' + y = 8e^x + x + 3$. [04]

(c) $x^2 y'' + xy' = 12 \log x$. [04]

(d) $y'' - 6y' + 9y = 0$ [02]

OR

Q-2 Solve the following differential equations

(a) $(x^4 - 2xy^2 + y^4)dx - (2x^2y - 4xy^3 + \sin y)dy = 0$. [04]

(b) $(2x+3)^2 y'' - 2(2x+3)y' - 12y = 6x$. [04]

(c) $\frac{dx}{dt} = 7x - y$, $\frac{dy}{dt} = 2x + 5y$. [04]

(d) $y'' + y' - 30y = 0$. [02]

Q-3 Solve the following Partial differential equations

(a) $(y-z)p + (z-x)q = (x-y)$. [04]

(b) (i) $p - x^2 = q + y^2$ (ii) $p^2 + q^2 = 4$. [04]

(c) Form the Partial differential equation (i) $2z = (x+a)(y+b)$ (ii) $z = xy + f(x^2 + y^2)$ [04]

(d) $(p+q)(z - px - qy) = 1$ [02]

OR

Q-3 Solve the following Partial differential equations

(a) $(mz - ny)p + (nx - lz)q = (ly - mx)$. [04]

(b) $p^2 z^2 + q^2 = 1$ [04]

(c) $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$ [04]

(d) $pqz = p^2(xq + p^2) + q^2(yp + q^2)$ [02]

SECTION-II

Q-4 (a) Evaluate $\iint_R dy dx$, R is the region bounded by $y^2 = 4x$ & $x^2 = 4y$. [03]

(b) If $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, Show that A is a Hermitian matrix and iA is a skew Hermitian matrix. [04]

Q-5 (a) Find the eigenvalues and corresponding eigenvectors for the matrix [05]

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

(b) Evaluate $\iint_R r^3 \sin 2\theta dr d\theta$ over the area bounded in the first quadrant R between the circles $r = 2$ and $r = 4$. [04]

(c) Show that $\int_0^\infty x^2 e^{-x^4} dx = \frac{1}{4} \sqrt{\frac{3}{4}}$. [03]

(d) (i) $\int_{-1}^1 P_3(x) P_4(x) dx = \underline{\hspace{2cm}}$ (ii) $\int_{-1}^1 P_6^2(x) dx = \underline{\hspace{2cm}}$. [02]

OR

Q-5 (a) State Cayley Hamilton theorem. Find the characteristic equation of the matrix [06]

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \text{ and hence evaluate the matrix equation}$$

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I.$$

(b) Prove $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$. [04]

(c) Change the order of integration in the integral $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ and evaluate it. [04]

Q-6 (a) Evaluate (i) $\int_0^\infty \frac{x^c}{c^x} dx$, ($c > 1$) (ii) $\int_0^n x^n (n-x)^p dx$. [06]

(b) Find the volume common to the cylinder $x^2 + y^2 = 4$ and the planes $y+z=3$ and $z=0$. [04]

(c) Express the function in $f(x) = x^3 + 2x^2 - x - 3$ in terms of Legendre polynomial. [04]

OR

Q-6 (a) Prove that (i) $P_n(1) = 1$ (ii) $P_n(-x) = (-1)^n P_n(x)$. [06]

(b) Prove that $\frac{1}{2} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$ is a unitary matrix. [04]

(c) Evaluate $\iiint_V z(x^2 + y^2) dx dy dz$ over the cylinder $x^2 + y^2 = 1$ intercepted by the plane $z=2$ and $z=3$. [04]